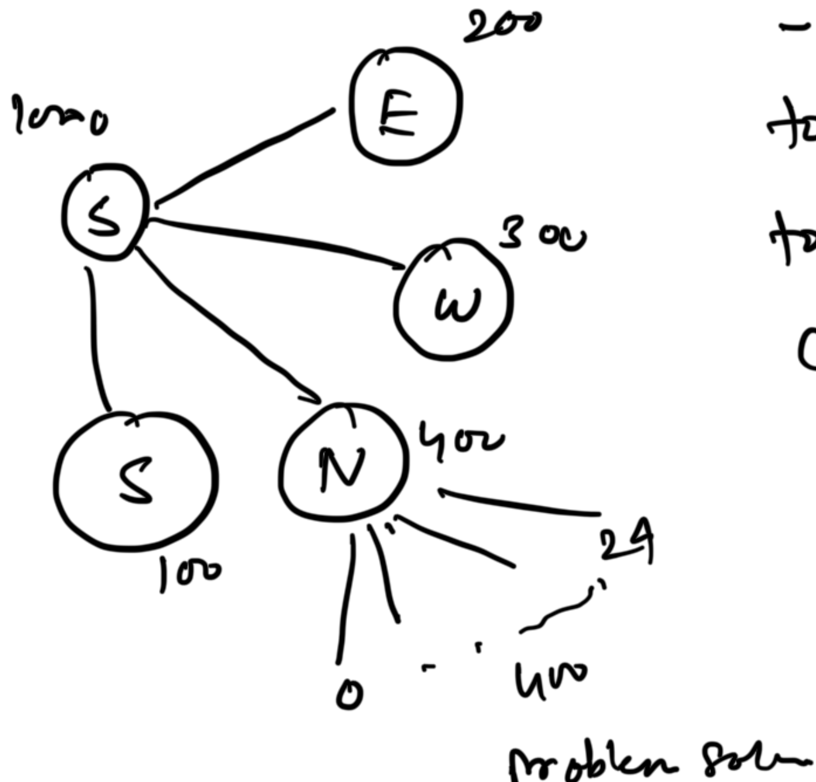
Spanning Tree & Shortest Path

Dynamic Programming - DP

- method for solving complex problems by breaking them into simpler problems.
- stores solutions to the subproblems to avoid redundant computations.



Greedy Method → every stage take a solution
→ Optimization of solution either min or max

\+

Dynamic Programming

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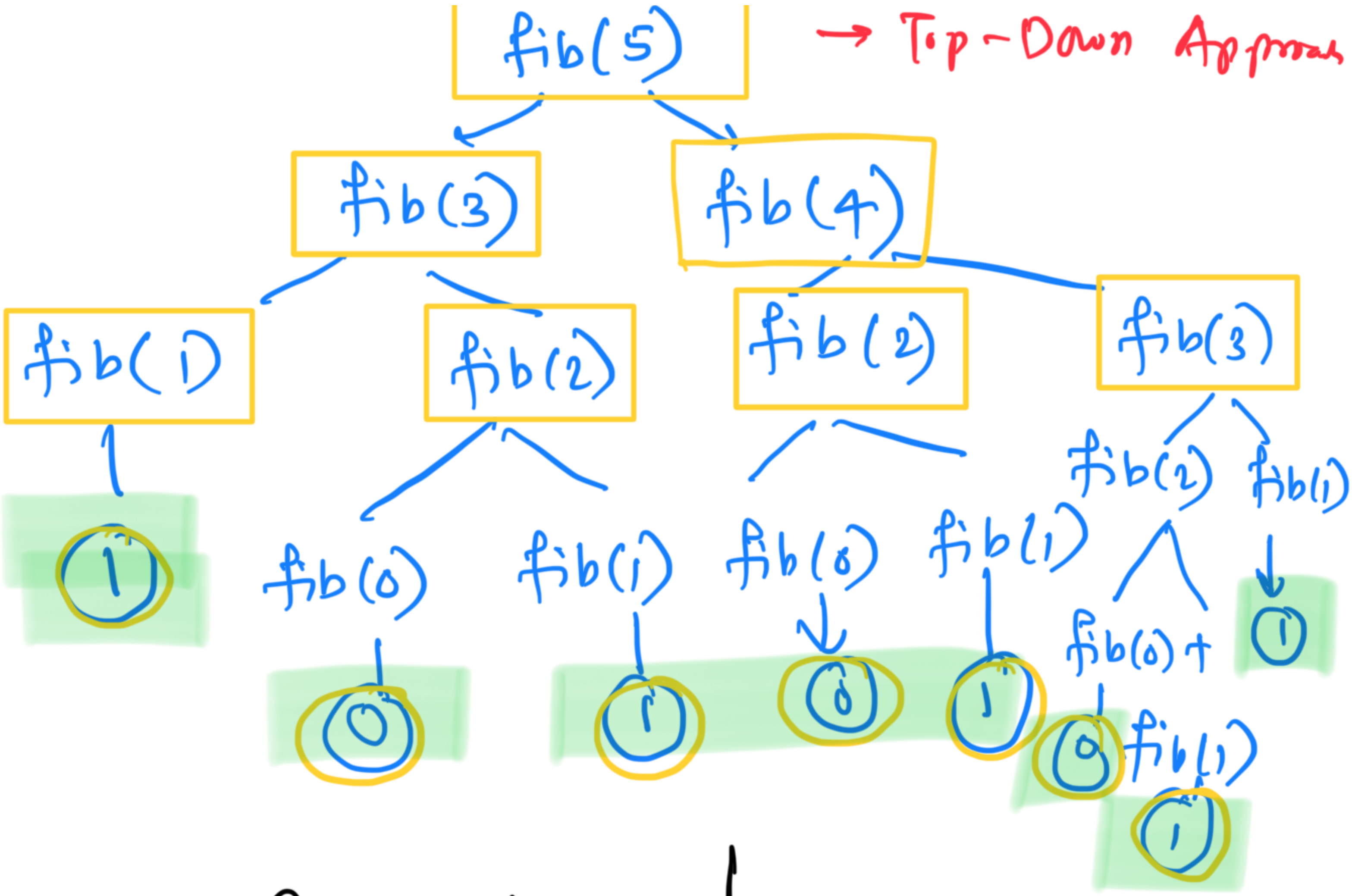
- Find all possible solⁿ then take best solution
- Recursion formula
- follow iteration
- Principal of optamility

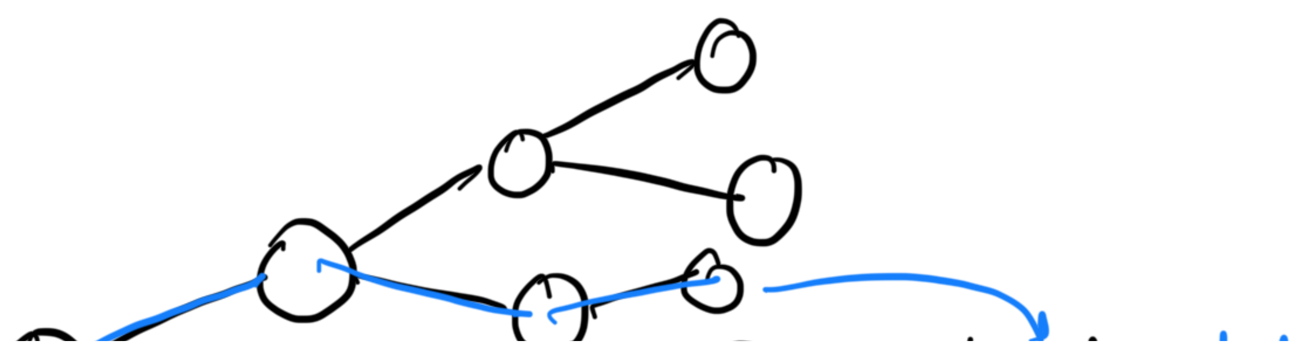
Ex: $$fib(n) = \begin{cases} 0 & \text{if } n=0 \\ 1 & \text{if } n=1 \\ fib(n-1) + fib(n-2) & \text{if } n>1 \end{cases}$$

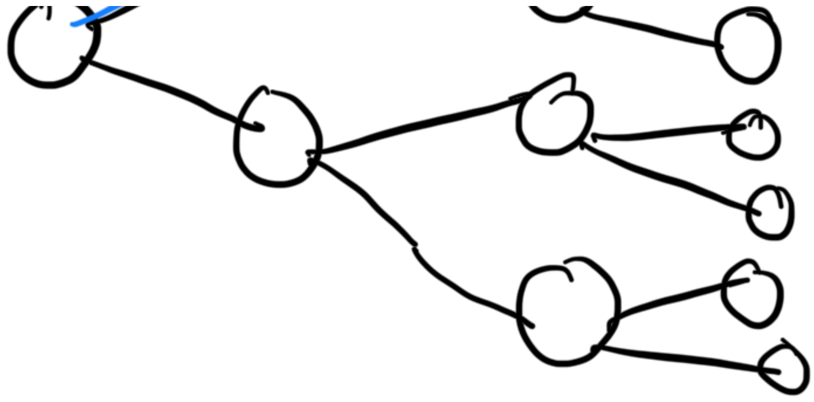
base cond^n (for $n=0$ and $n=1$)

→ Top-Down Approach



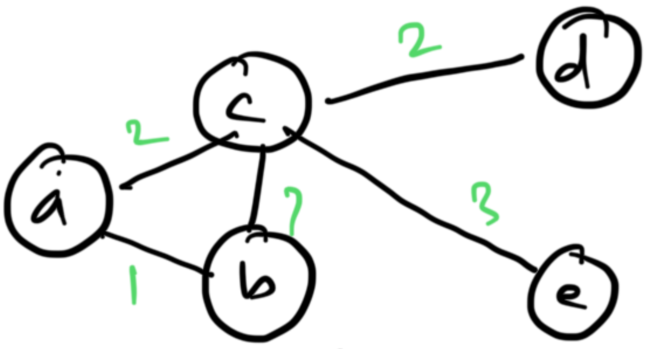
Greedy Approach	Dynamic programming
best option that gives us the profit	Optimized solⁿ recursion, backtracking
Not necessary to be globally optimal	Gives optimal solⁿ
Ex. Prim's Algo Dijkstra Algo	Ex. 0/1 knapsack Longest Increasing Subsequence
Top-down Approach	Bottom-Up Approach





optimal solution

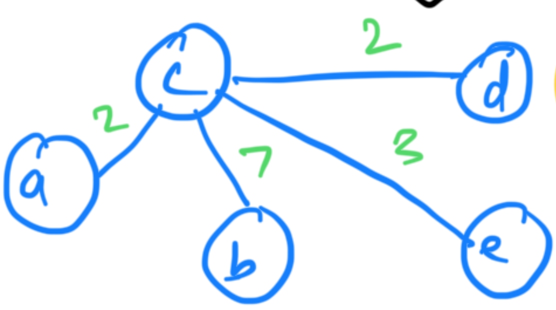
Spanning Tree



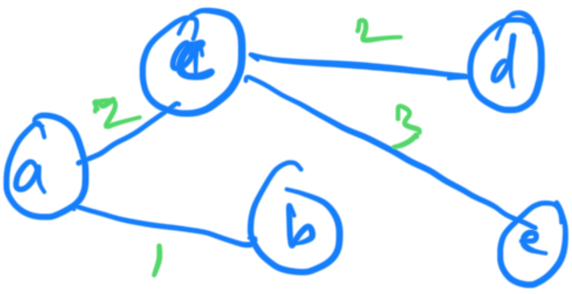
G(V,E)

⇓

A subgraph of a undirected graph G (V,E) is spanning tree of G, if it is a tree and must contains vertex G

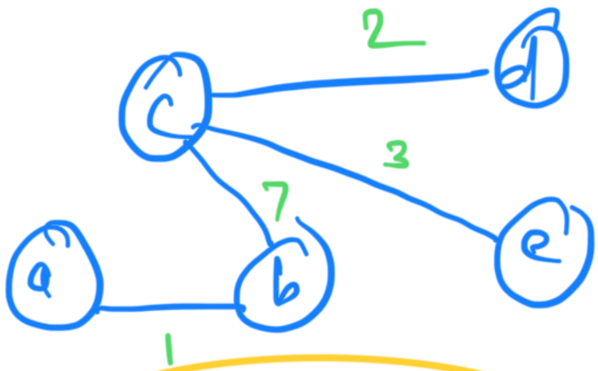


SP1 = 14



SP2 = 8

Min Spanning Tree



SP 3 = 13

No of vertices = 5

Edges = 5

ST

V = 5

Edges = n-1
= 4

How many ST can be generated :)

$${}^{n}C_{E} = {}^{5}C_{4} = \ ?$$

Spanning Tree :

Minimum Spanning Tree - (MST)

- Among all the spanning tree of a graph, the MST is the one with smallest possible total cost.

Conditions-

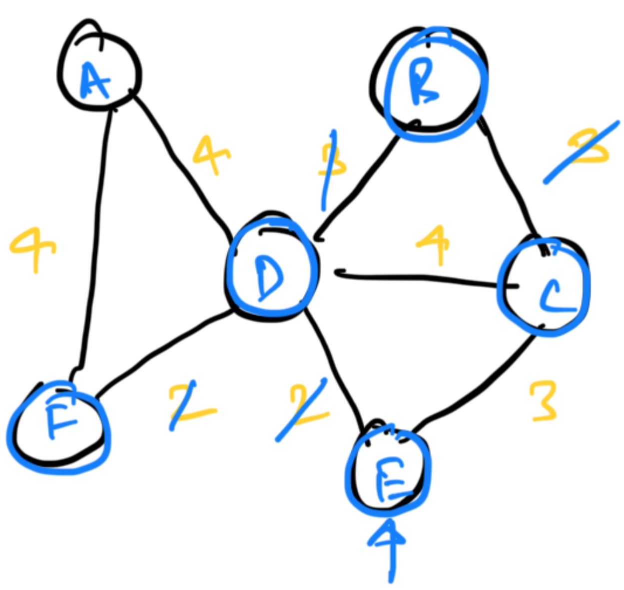
1. The graph must be connected
2. The graph must be undirected
3. If all edge weights are unique, the MST is unique.

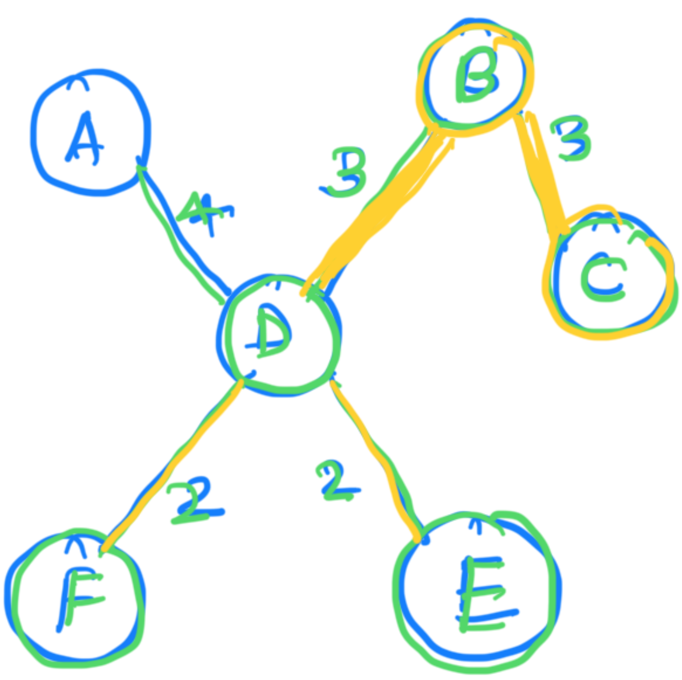
Algorithms for minimum cost Spanning tree

1. Krushkal's Algo
2. Prim's Algo

} Greedy Algorith Strategy

Krushkal's Algorithm





MST cost = 14

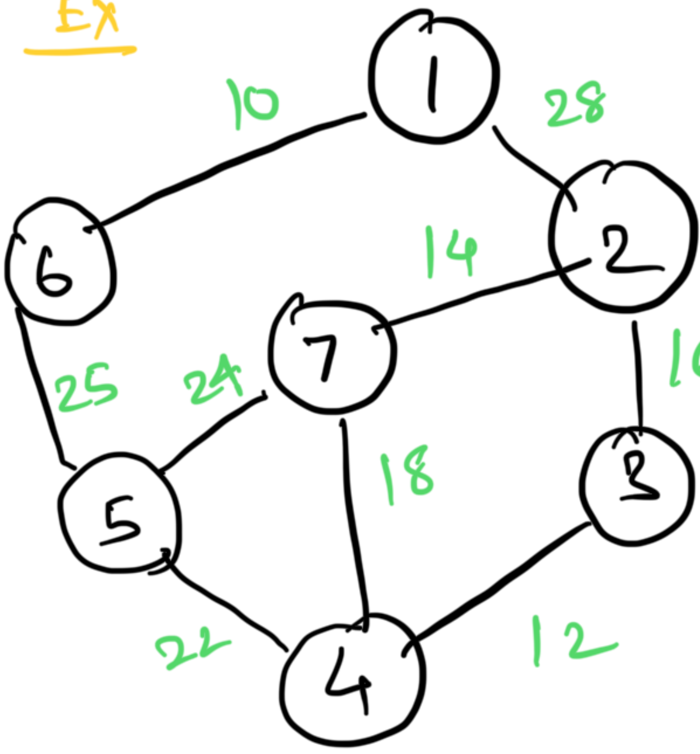
- Greedy Algorithm

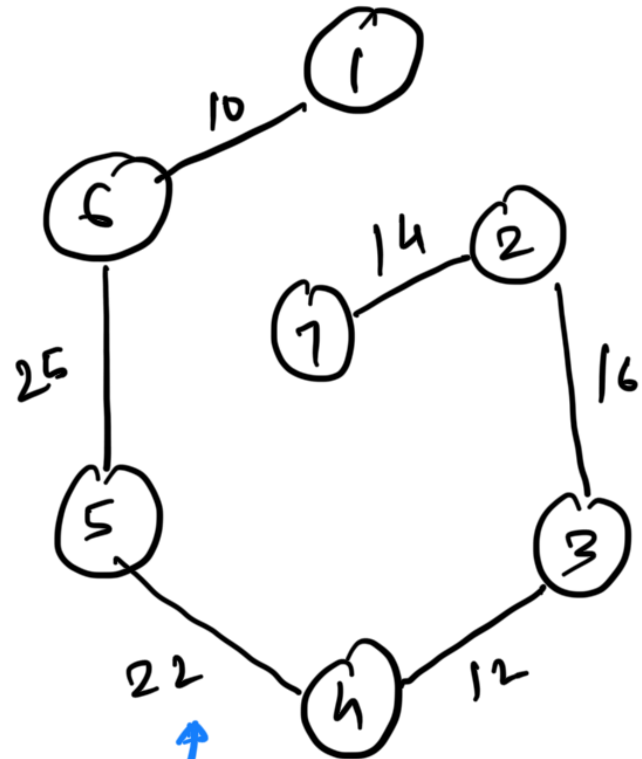
- MST by sorting all the edges and adding them one by one, ensuring that no cycle is formed

Edge ⇒ minimum cost → selection criteria

2, 2, 3, 3, 4, 4, 4

Ex





sorted edges.

10, 12, 14, 16, 22, 25 ⇒ MST = 99

Time Complexity = $O(E \log E)$

E ⇒ no. of edges

Space Complexity = $O(E + V)$

Edges = $|E|$

Vertices = $|V|$

Time Comp ⇒ $O(|V| \cdot |E|)$

$O(n \times e)$

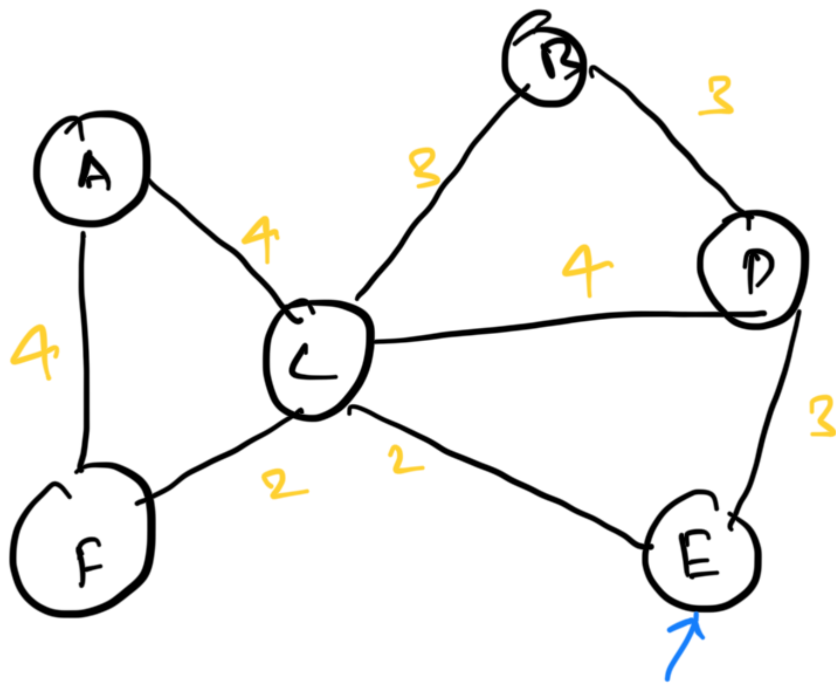
$\approx O(n \times n) \approx O(n)$

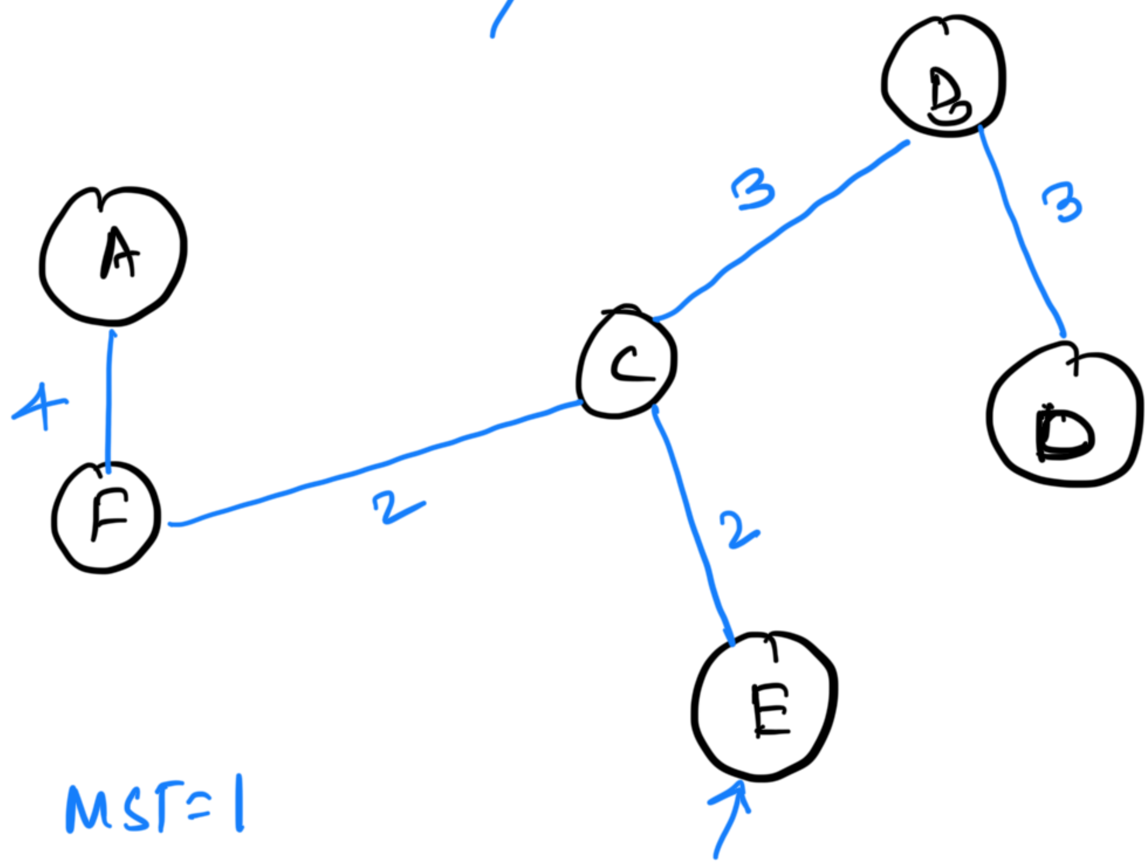
Improve Time Complex $\Theta(E \log E)$

Prim's Algorithm

- Greedy algorithm
- MST by starting from a single node & repeatedly adding the smallest edge that connects a vertex inside the tree to the vertex outside the tree

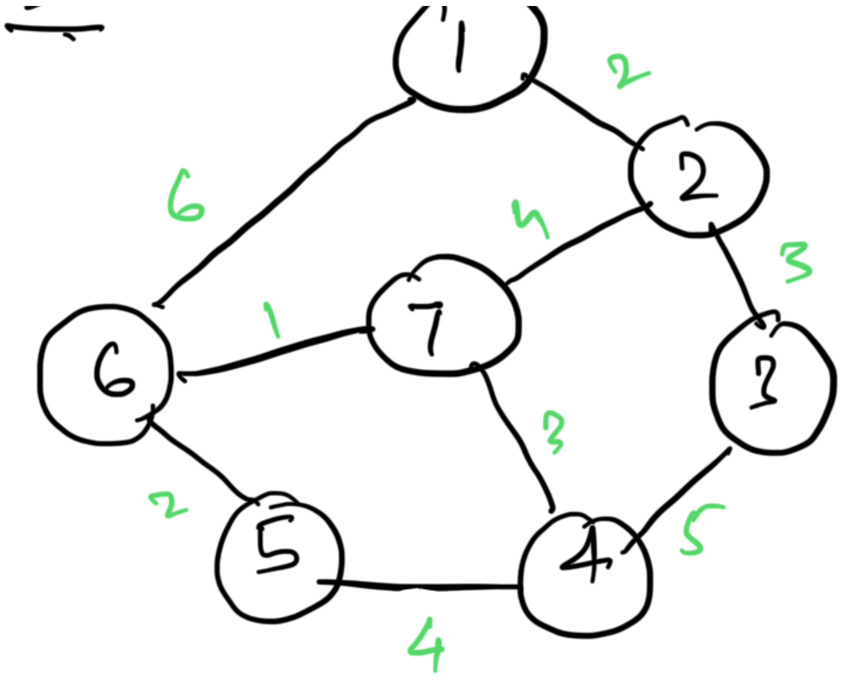
Ex:

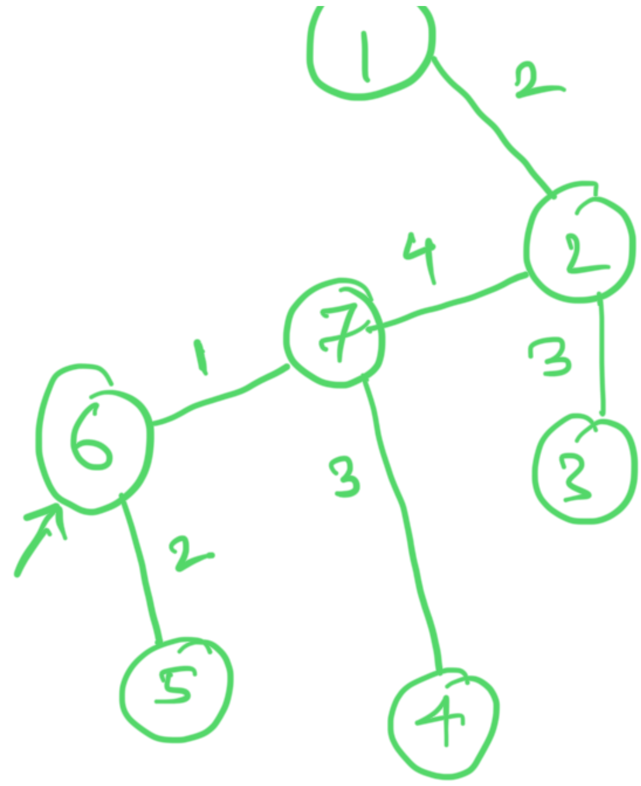




MST = 1

Ex:





MST = 15

Time Complexity ⇒ $O(E \log V)$

Space Complexity ⇒ $O(V)$

Shortest Path Algorithm: Dijkstra's Algorithm

→ Greedy Algorithm

→ minimum cost

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optimization problem

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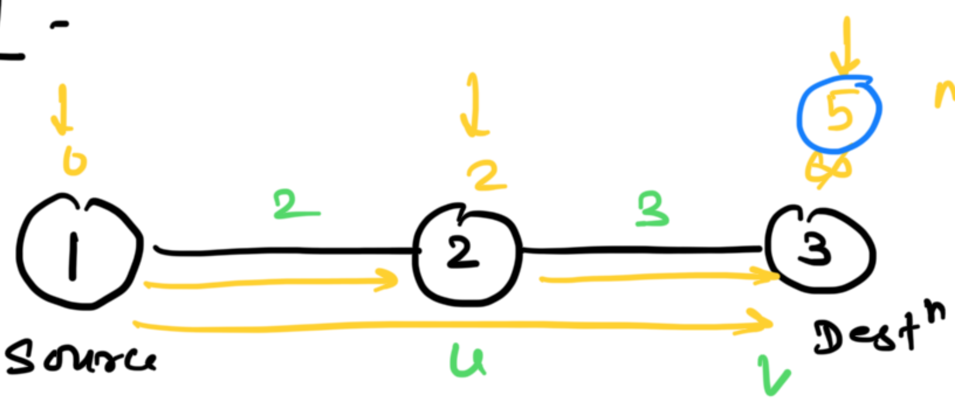
Minimization problems. ⇒ $d(u, v)$

source dest

<u>Method</u> -



minimum directed path

$2+3 < \infty$

$5 < \infty$

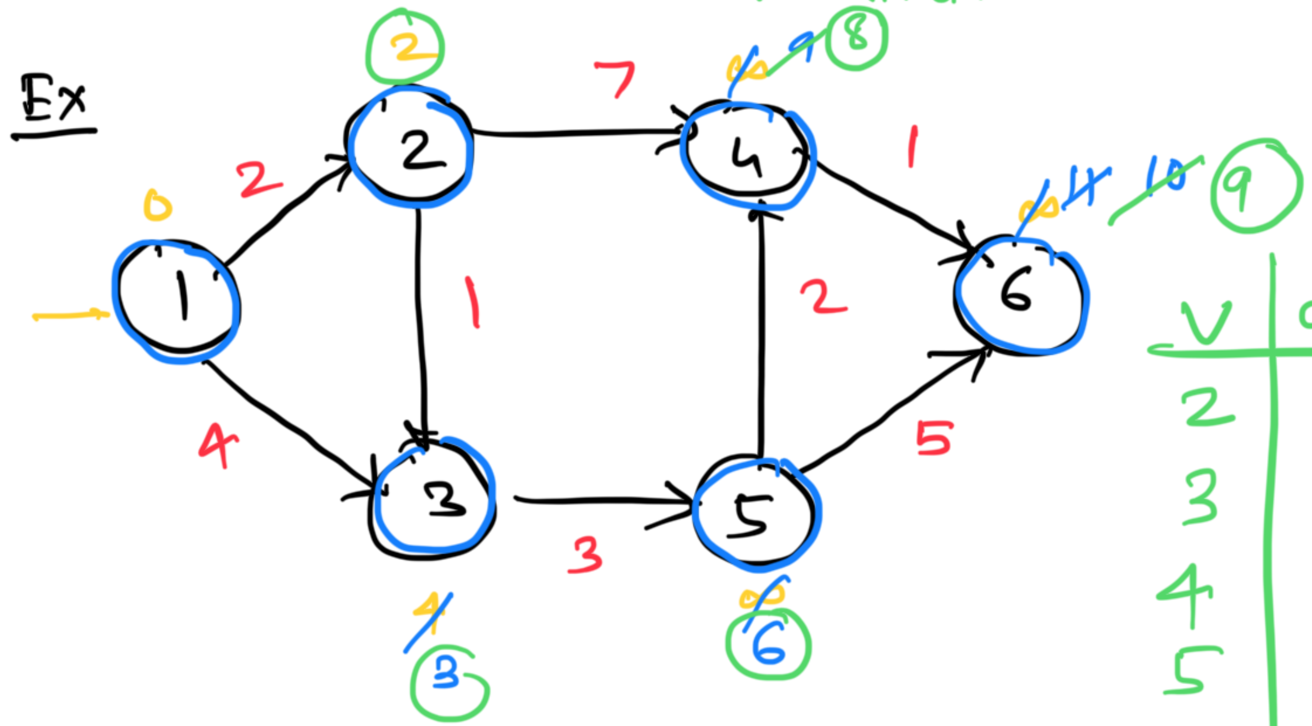
→ update the directed path to shortest vertices. ⇒ $\boxed{\text{Relaxation}}$

$$\text{if } \left(d[u] + c[u,v] < d[v] \right)$$

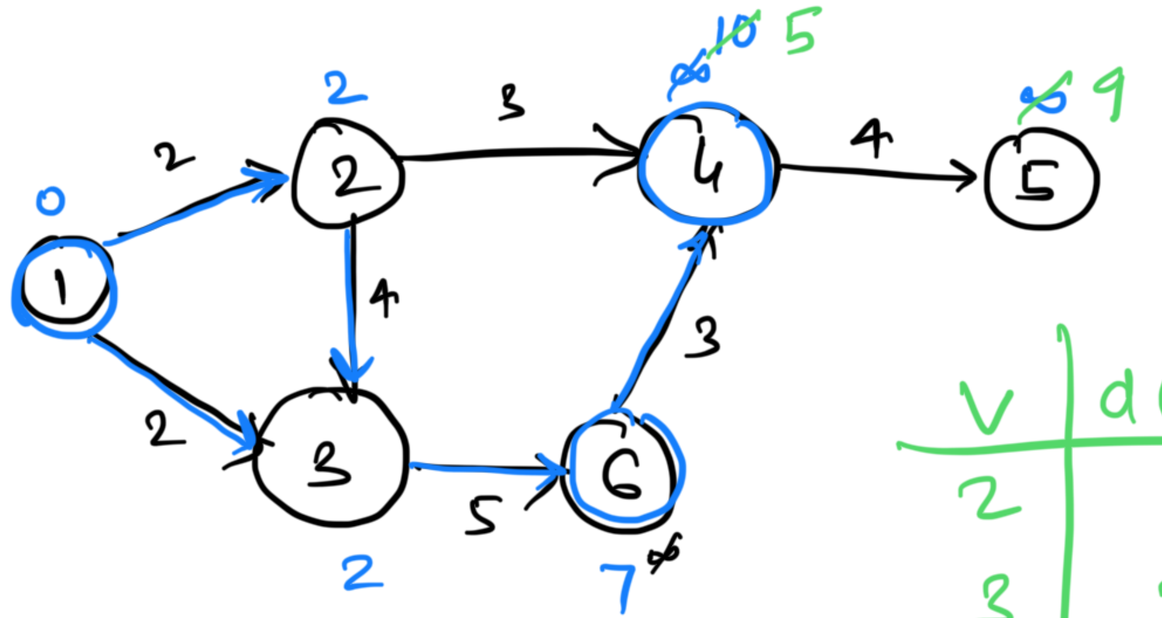
$$\boxed{d[v] = d[u] + c[u,v]}$$

Relaxation

Ex



v	d(v)
2	2
3	3
4	8
5	6
6	9



v	d(v)
2	2
3	2
4	5
5	9
6	7

Time Complexity $= n \times n$

$= O(n^2)$

$= O(|V|^2)$

Summary -

Prims Algo $\rightarrow O(E \log V)$

Krushkal Algo $\rightarrow O(E \log E)$

Dijakstra Algo $\rightarrow O(|V|^2)$

Bellman-Ford Algo $\rightarrow O(|E| \cdot |V|)$

Flody Warshall's Algo $\rightarrow O(|V|^3)$